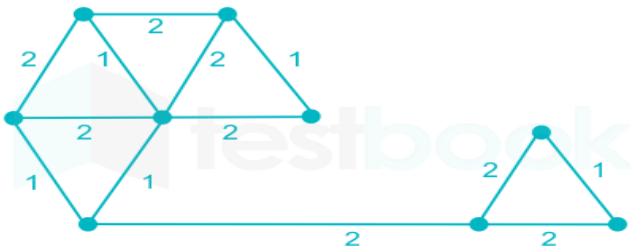
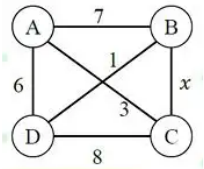
Assignment-II

1. Find the absolute minimum value of the function $f(x, y) = x^2 + y^2$ subject to the constraint $x^2 + 2y^2 = 1$.
2. A firm manufactures a commodity at two different factories. The total cost of manufacturing depends on the quantities, p and q supplied by each factory, and is expressed by the joint cost function $2pq^2 + pq + 22q + 500$. The company's objective is to produce 200 units, while minimizing production costs. How many units should be supplied by each factory?
3. Find the maximum possible weight of the minimum spanning tree of a graph with four vertices and six edges having weights $6, 7, 4, 8, 11, 9$.
4. A manufacture produces two types of product x and y. The profit is given by $2x + 3y$. What will be the maximum profit if the constrains are $x + 3y \leq 40, 3x + y \leq 24, x + y \leq 10, x > 0, y > 0$. Use graphical method.
5. Use graphical method to solve the following LPP. $Max\ Z = 3x + 2y$ Subject to $5x + y \geq 10,\ x + y \geq 6,\ x + 4xy \geq 12,\ x, y \geq 0$.
6. Use graphical method to solve the following LPP. $Max\ Z = x + y$ Subject to $x + y \leq 10,\ x + y \geq 3,\ x, y \geq 0$.
7. Use simplex method to solve the following LPP. $Max\ Z = 3x + 2y$ Subject to $x + 3y \leq 40, 3x + y \leq 24, x + y \leq 10, x > 0, y > 0$

8.

The number of distinct minimum spanning trees for the weighted graph below is _____



9.

The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is _____.